CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture M4a: Point Estimates, Sampling Variability and Central Limit Theorem

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Outline

1. Statistical inference
2. Point estimation
3. Method of moments
4. Variability and CLT
5. Outlook

Statistical inference

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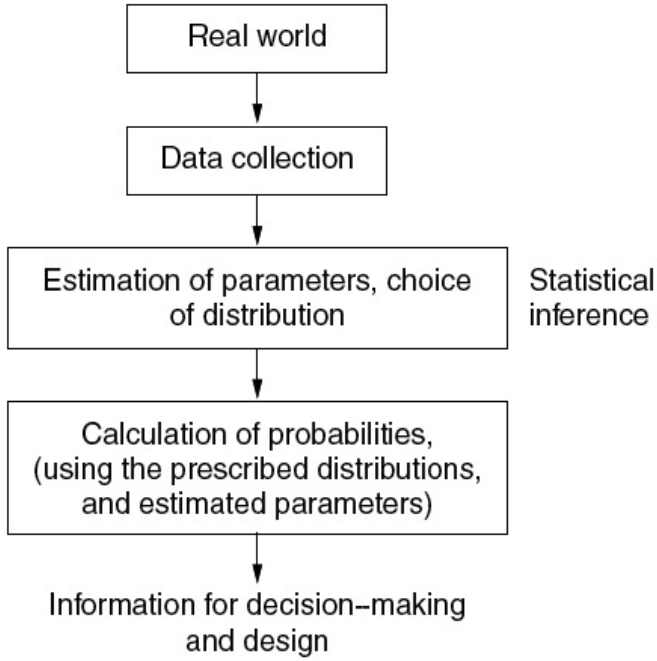
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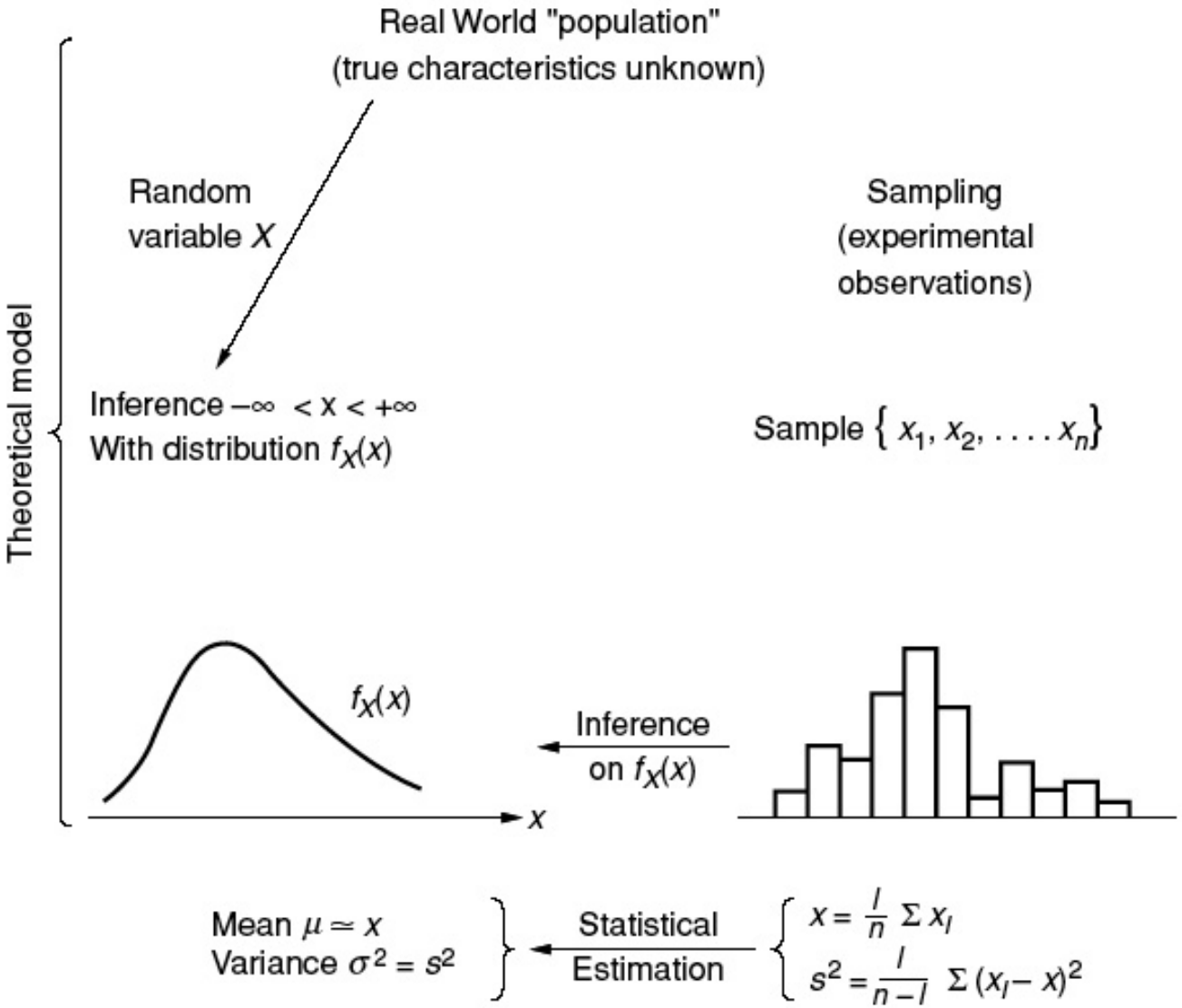
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Role of sampling in statistical inference



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Definition

A **point estimate**, $\hat{\theta}$, of a parameter θ (e.g. proportion p, or mean value μ) is a single number that can be regarded as a sensible value for θ and is obtained by computing the value of a suitable statistic (e.g. sample mean, sample standard deviation, etc) from given sample data.

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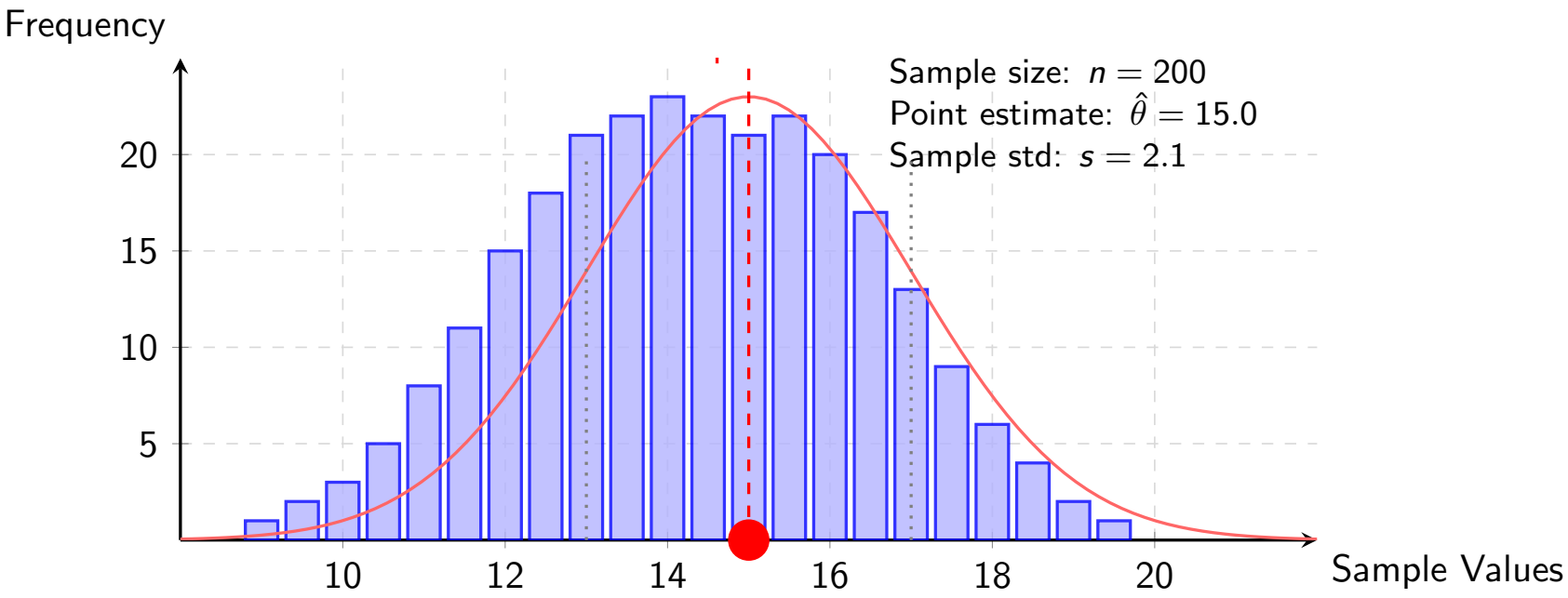


Figure: Sample histogram with point estimate $\hat{\theta}$ showing the center of the distribution

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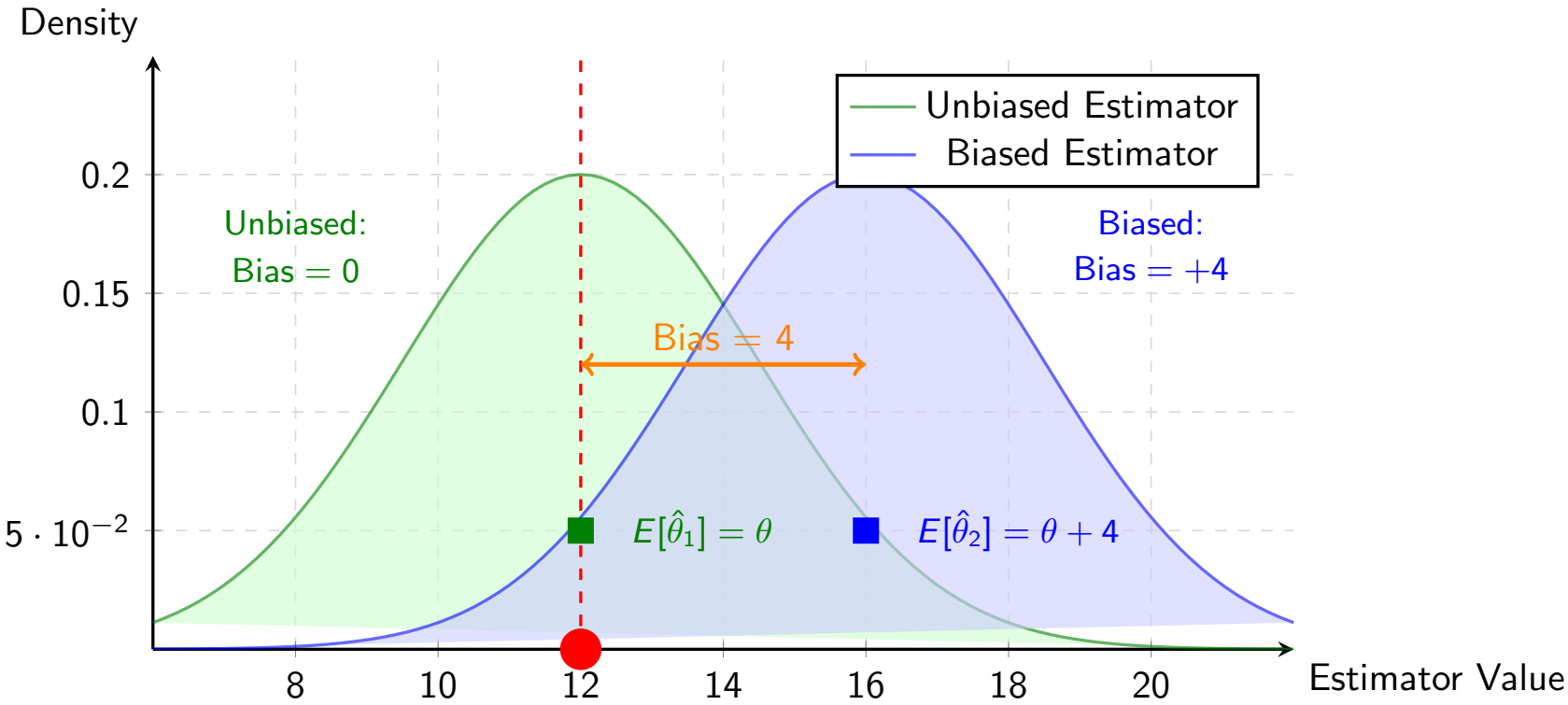
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An estimator is *consistent* if $\hat{\theta} \to \theta$ as $n \to \infty$,i.e. the estimation error should decrease with increasing sample size.

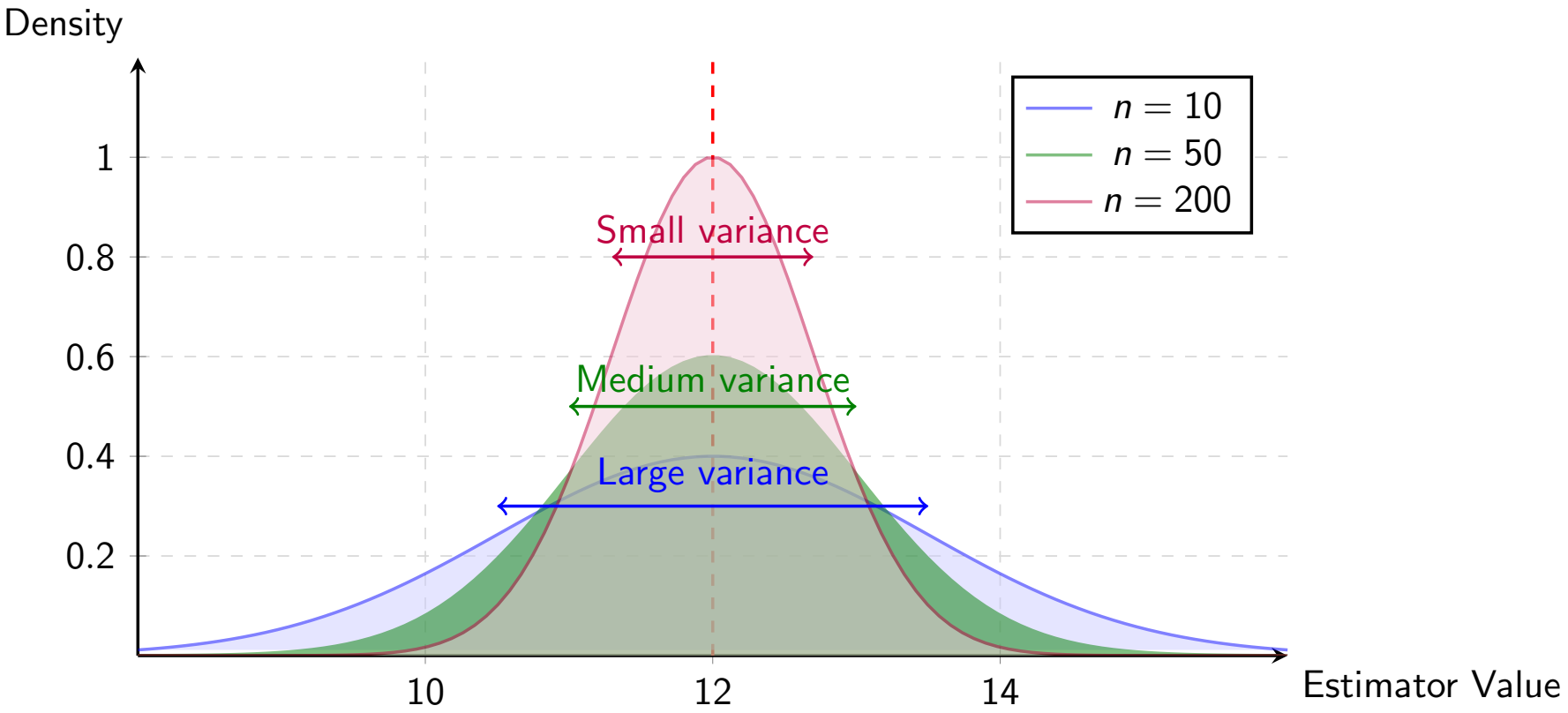


Figure: As sample size increases, the sampling distribution becomes more concentrated around the true parameter

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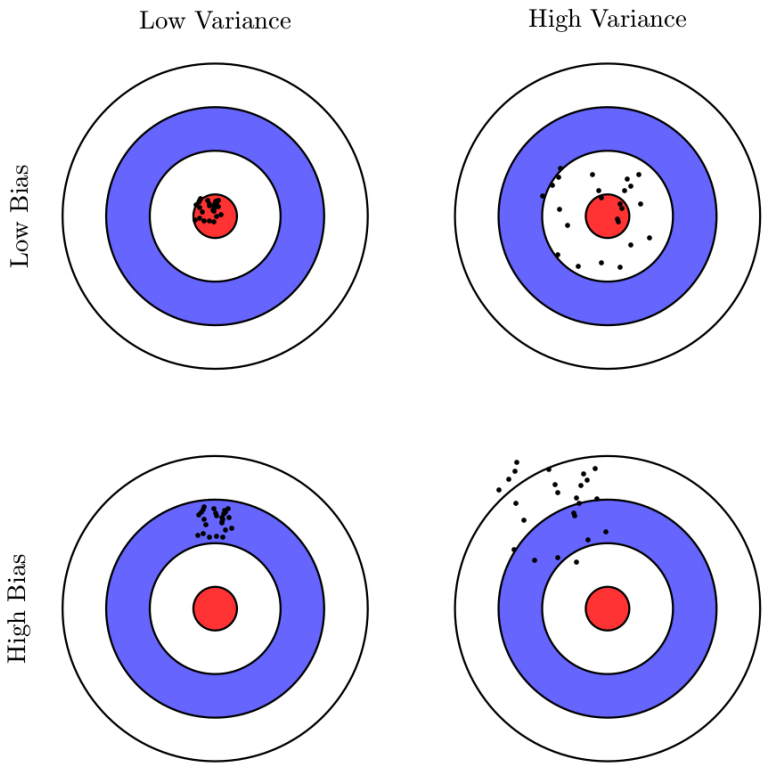


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$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2 \tag{5}$$

Unbiasedness of s^2

From Equation (5), you can show (as an exercise) that:

$$s^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - n\overline{x}^2\right] \tag{6}$$

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(c) Now, estimate the variance replacing the denominator $(n-1)$ with n in the estimator s^2. What do you notice?

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Example 1: Elastic modulus of alloys (cont.)

$X = 44.2, 43.9, 44.7, 44.2, 44.0, 43.8, 44.6, 43.1$

(a) $\hat{\mu} = \overline{x} = \frac{1}{8}\sum_{i=1}^{8} x_i \approx \boxed{44.063}$

(b) $s^2 = \frac{1}{7}\left[\sum_{i=1}^{8} x_i^2 - 8(44.063^2)\right] \approx \boxed{0.251}$

(c) Biased estimate of σ^2:

$\hat{\sigma}^2 = \frac{1}{8}\left[\sum_{i=1}^{8} x_i^2 - 8(44.063^2)\right] = \frac{7}{8}(0.251) = \boxed{0.220}$

$\hat{\sigma}^2$ underestimates σ^2 by 0.031 squared units.

The Central Limit Theorem (CLT)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. If n is sufficiently large, then the sample mean $\overline{X}$ has approximately a **normal distribution** with

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- The product of large number of random components approaches the lognormal distribution

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Sample proportion and the CLT

If the observations in a given sample are a Bernoulli sequence with a constant proportion (or probability) p, then if n is large, the sample proportion $\hat{p}$ follows a normal distribution (according to the CLT):

$$\hat{p} \sim \mathcal{N}(\mu_{\hat{p}}, SE_{\hat{p}}^2) = \mathcal{N}\left(p, \frac{p(1-p)}{n}\right) \tag{17}$$

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One rule of thumb for determining whether n is large enough is to check that both np and $n(1-p)$ are ≥ 10 (also known as the success-failure condition).

CLT application: sample proportion

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Example 2: Solar energy expansion (CLT)

Suppose the proportion of American adults who support the expansion of solar energy is $p = 0.88$, which is our parameter of interest.

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(a) According to the CLT, what is the distribution of $\hat{p}$?

(b) According to the CLT, what are $\mu_{\hat{p}}$ and $SE_{\hat{p}}$, respectively?

CLT application: sample proportion (cont.)

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Example 3: Solar energy expansion (CLT)

(a) First, we note that the response of each American adult in the entire population is part of a Bernoulli sequence with $p = 0.88$. According to the CLT, the distribution of $\hat{p}$ (sample proportion) is normal/Gaussian. We can denote this as:

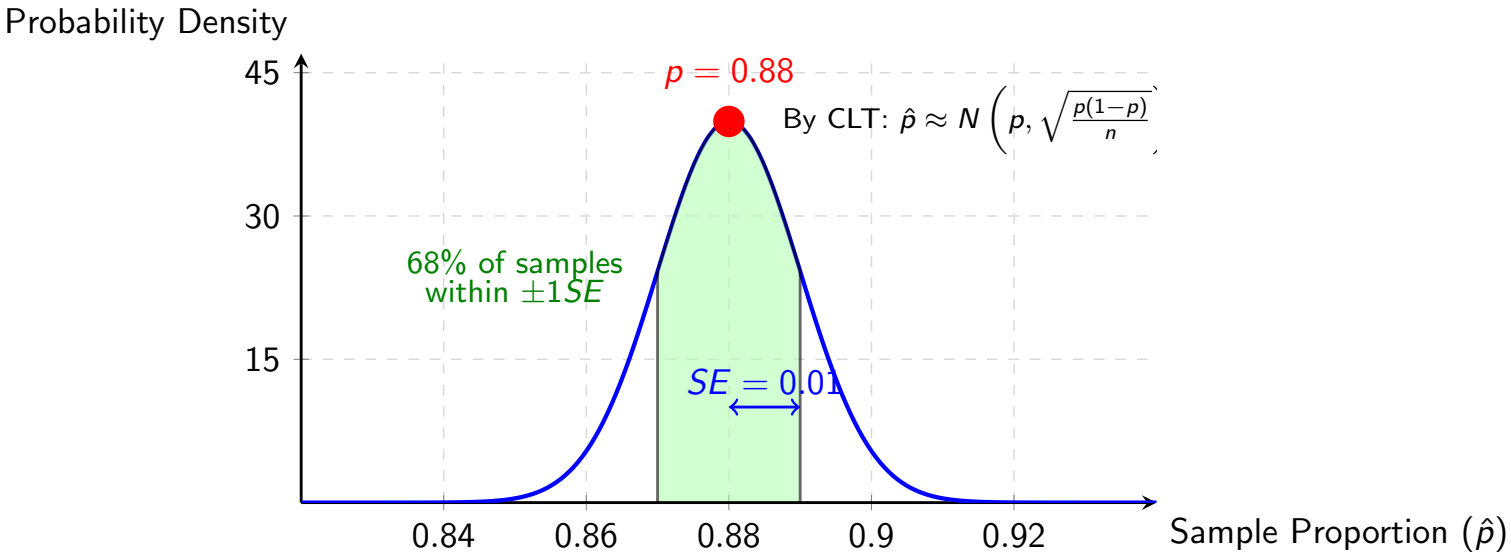
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(b) $\mu_{\hat{p}}$ denotes the mean estimate of p, which is 0.88 (according to the CLT, the mean of the sample is the population mean if n is large).
$SE_{\hat{p}}$ denotes the sampling error, which is the the square root of the variance of the sample mean: $\sqrt{\sigma^2/n}$. Given that the sample is governed by the Binomial distribution with $\sigma^2 = p(1-p)$. Thus:

$$SE_{\hat{p}}^2 = \frac{\sigma^2}{n} = \frac{p(1-p)}{n} = \frac{0.88(0.12)}{1000}$$

$$SE_{\hat{p}} = \sqrt{\frac{0.88(0.12)}{1000}} = \boxed{0.01}$$

CLT application: sample proportion (cont.)

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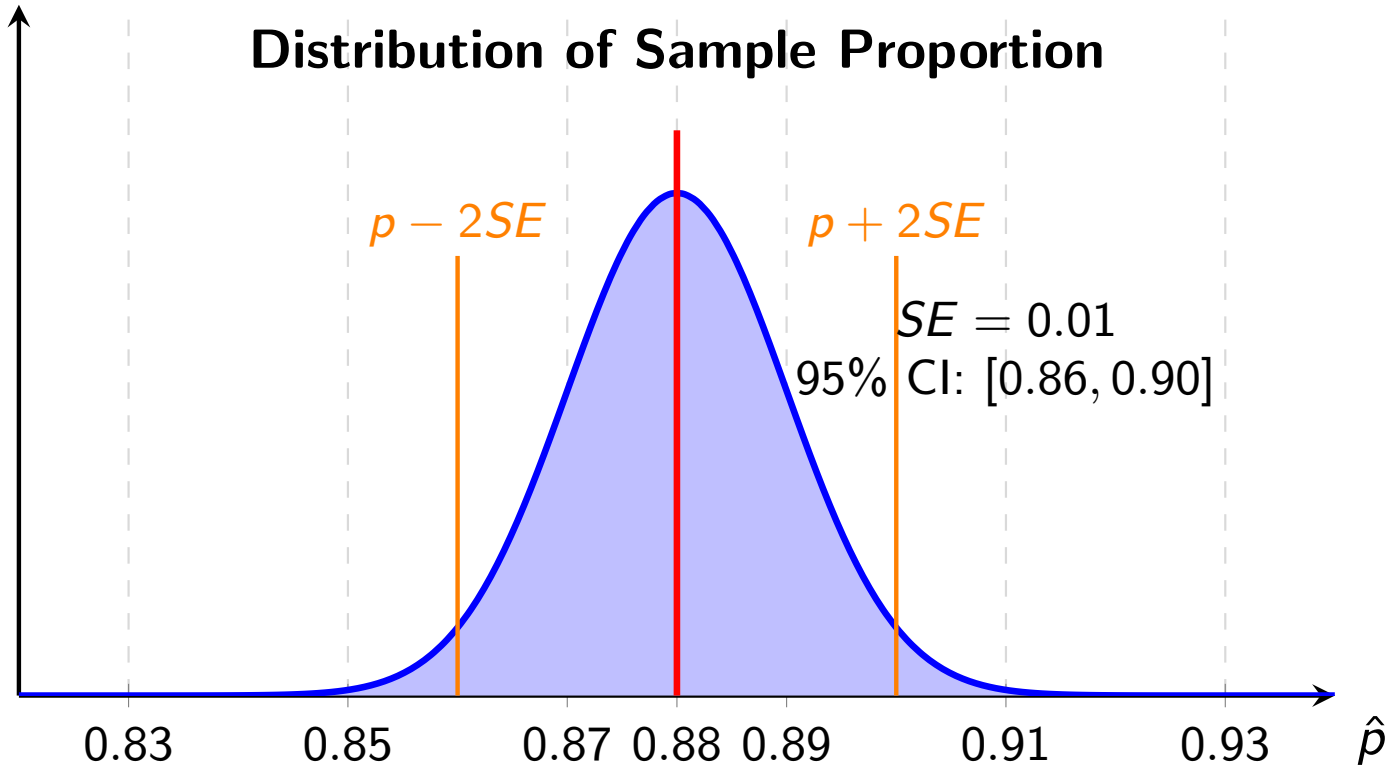


Figure: Sample proportion distribution: most samples fall within $\pm 2SE$ of the true proportion

Another application of the CLT

Example 3: Mean batch weight

A certain brand of cement is shipped in batches of 40 bags. Previous records indicate the weight of a randomly selected bag of this brand has a mean of 2.5 kg and an SD of 0.1 kg. The exact distribution is unknown.

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(b) If the shipping company charges an overweight fee if a batch exceeds the mean batch weight by more than 1 kg, what is the probability that a batch will be charged?

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$$n(1-p) \geq 10 \tag{21}$$

Summary

- Desired properties of point estimates: unbiasedness and efficiency
- Distribution of sample proportions (or other parameters) is called a sampling distribution
- When n is sufficiently large and observations are independent, the sample proportion (or other parameter) follows a normal distribution
- The success-failure condition can be used to determine if n is large enough for the CLT to hold (for a sample proportion)

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